



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

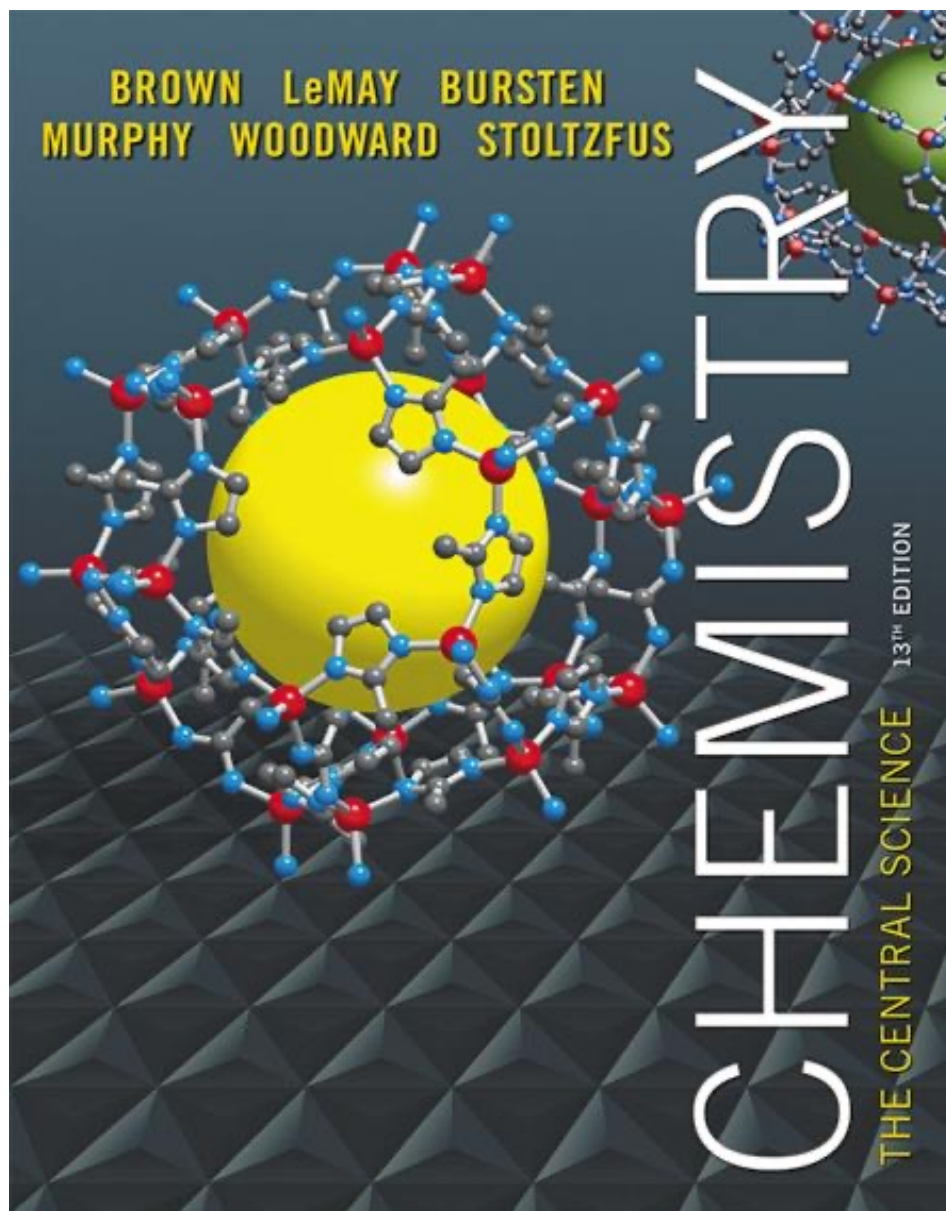
General Chemistry CHM1001

Chemical Equilibrium (Ch.15)

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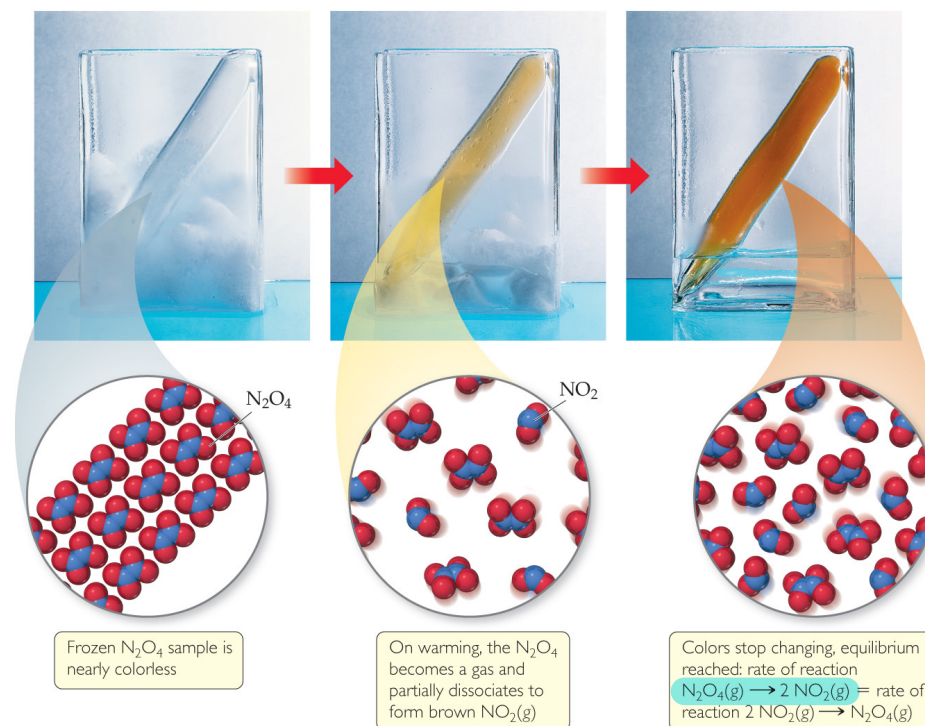
Lecture Presentation

Chapter 15

Chemical Equilibrium

Aqueous
Equilibria

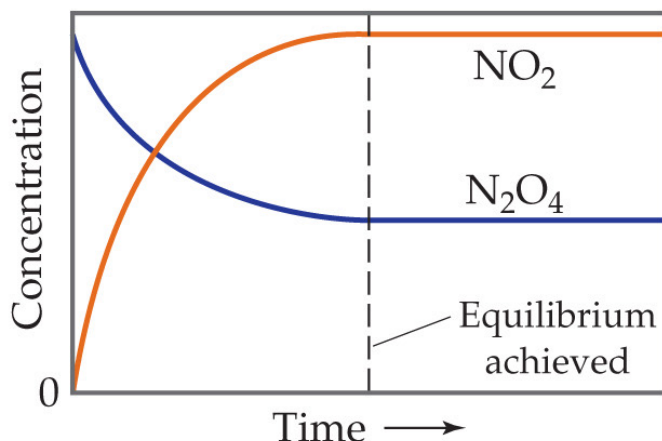
The Concept of Equilibrium



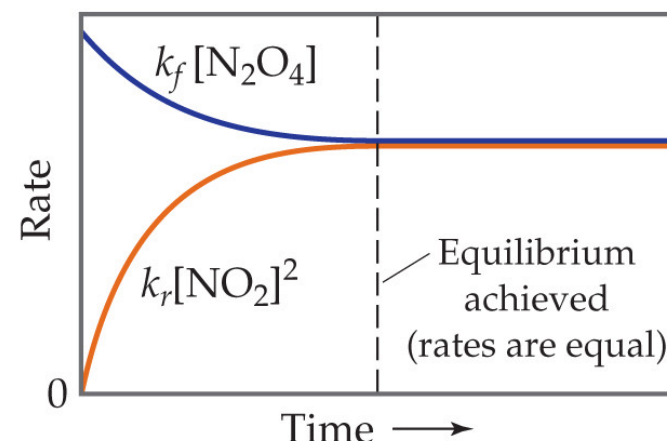
Chemical equilibrium occurs when a reaction and its reverse reaction proceed at the same rate. In the figure above, equilibrium is finally reached in the third picture.

Aqueous
Equilibria

The Concept of Equilibrium



(a)



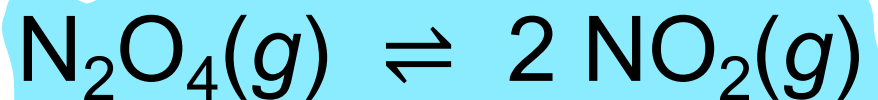
(b)

- As a system approaches equilibrium, both the forward and reverse reactions are occurring.
- At equilibrium, the forward and reverse reactions are proceeding at the same rate.
- Once equilibrium is achieved, the amount of each reactant and product remains constant.



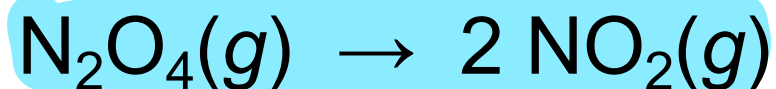
Writing the Equation for an Equilibrium Reaction

Since, in a system at equilibrium, both the forward and reverse reactions are being carried out, we write its equation with a double arrow:



Comparing Rates

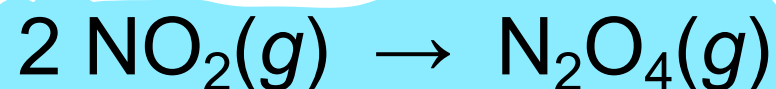
- For the forward reaction



- The rate law is

$$\text{Rate} = k_f[\text{N}_2\text{O}_4]$$

- For the reverse reaction



- The rate law is

$$\text{Rate} = k_r[\text{NO}_2]^2$$



The Meaning of Equilibrium

- Therefore, at equilibrium

$$\text{Rate}_f = \text{Rate}_r$$

$$k_f[\text{N}_2\text{O}_4] = k_r[\text{NO}_2]^2$$

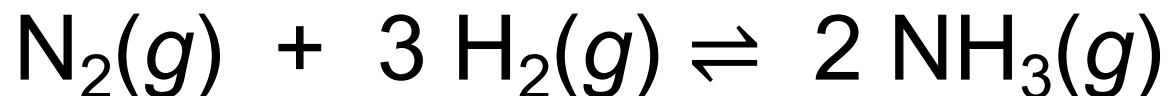
- Rewriting this, it becomes the expression for the equilibrium constant, K_{eq} .

$$\frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{k_f}{k_r} = \text{a constant}$$



Another Equilibrium— The Haber Process

- Consider the Haber Process, which is the industrial preparation of ammonia:



- The equilibrium constant depends on stoichiometry:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Product
Reactant



The Equilibrium Constant

- Consider the generalized reaction



- The equilibrium expression for this reaction would be

$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

← products
← reactants

- Also, since pressure is proportional to concentration for gases in a closed system, the equilibrium expression can also be written

$$K_p = \frac{(P_D)^d (P_E)^e}{(P_A)^a (P_B)^b}$$

More with Gases and Equilibrium

- We can compare the equilibrium constant based on concentration to the one based on pressure.
- For gases, $PV = nRT$ (the Ideal Gas Law).
- Rearranging, $P = (n/V)RT$; (n/V) is ^{molar} [].
- The result is

$$K_p = K_c (RT)^{\Delta n}$$

- where

$$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$$

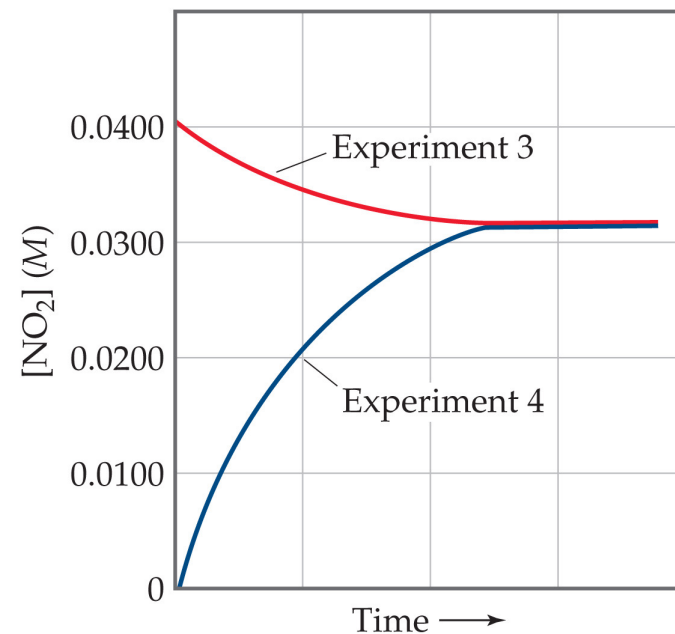


Equilibrium Can Be Reached from Either Direction

- As you can see, the ratio of $[\text{NO}_2]^2$ to $[\text{N}_2\text{O}_4]$ remains constant at this temperature no matter what the initial concentrations of NO_2 and N_2O_4 are.

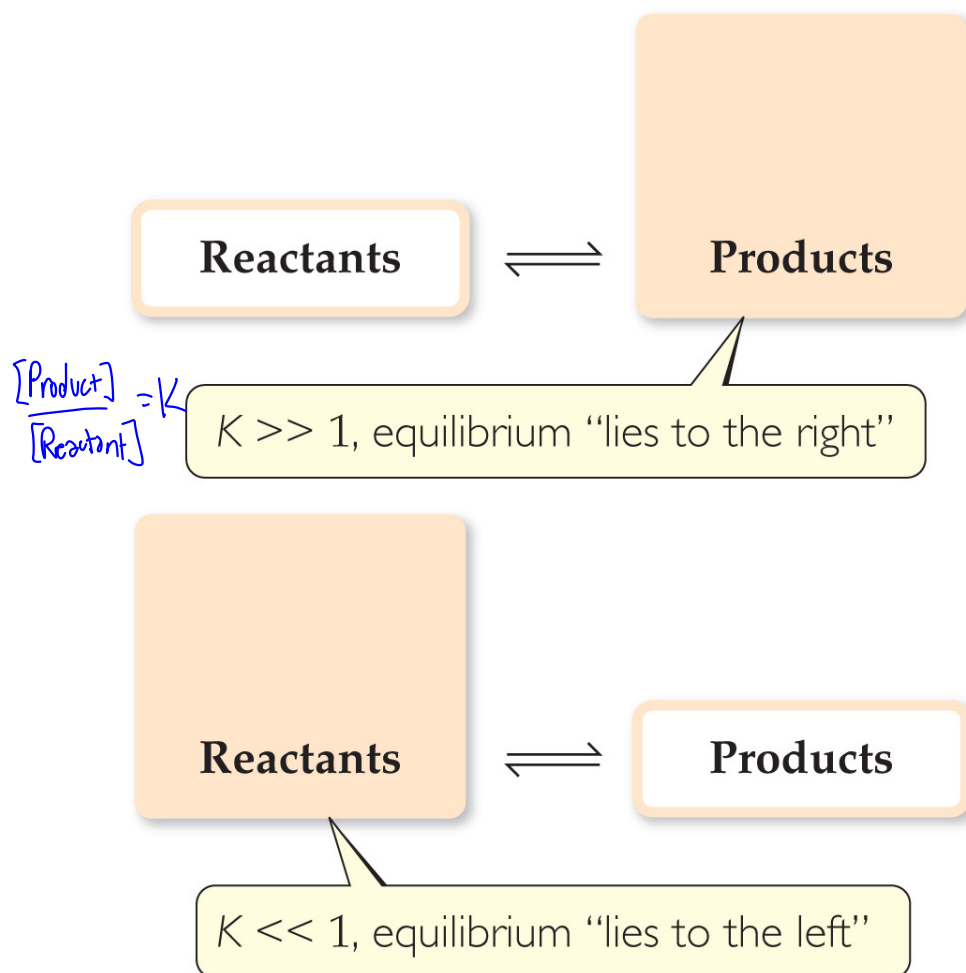
Table 15.1 Initial and Equilibrium Concentrations of $\text{N}_2\text{O}_4(g)$ and $\text{NO}_2(g)$ at 100°C

Experiment	Initial $[\text{N}_2\text{O}_4]$ (M)	Initial $[\text{NO}_2]$ (M)	Equilibrium $[\text{N}_2\text{O}_4]$ (M)	Equilibrium $[\text{NO}_2]$ (M)	K_c
1	0.0	0.0200	0.00140	0.0172	0.211
2	0.0	0.0300	0.00280	0.0243	0.211
3	0.0	0.0400	0.00452	0.0310	0.213
4	0.0200	0.0	0.00452	0.0310	0.213



Aqueous
Equilibria

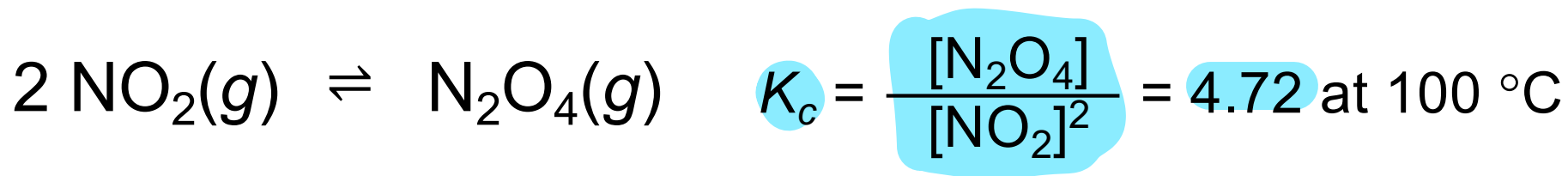
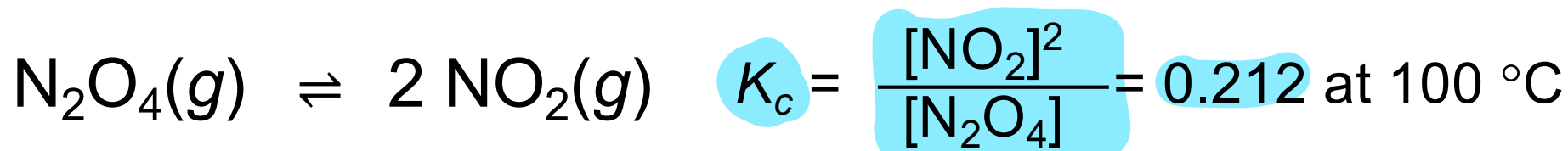
Magnitude of K



- If $K \gg 1$, the reaction favors products; products predominate at equilibrium.
- If $K \ll 1$, the reaction favors reactants; reactants predominate at equilibrium.

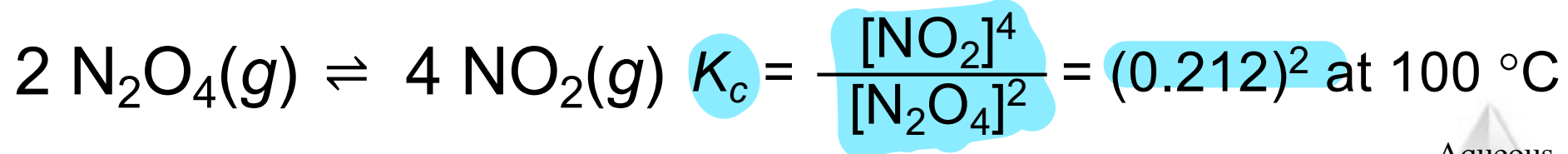
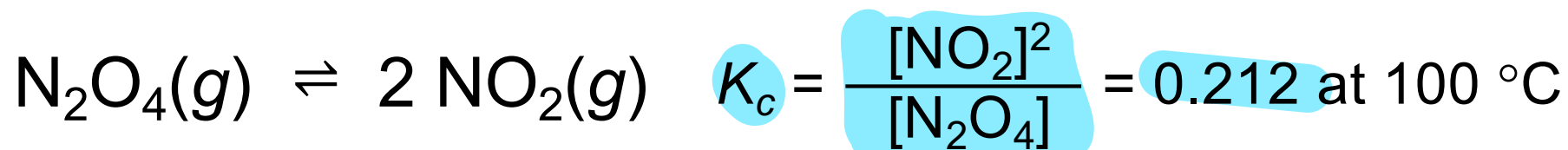
The Direction of the Chemical Equation and K

The equilibrium constant of a reaction in the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction:



Stoichiometry and Equilibrium Constants

To find the new equilibrium constant of a reaction when the equation has been multiplied by a number, simply raise the original equilibrium constant to that power. Here, the stoichiometry is doubled; the constant is the squared!



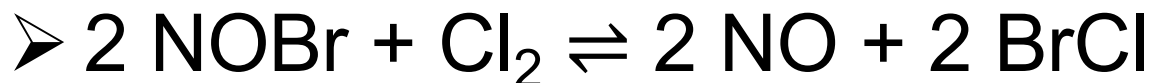
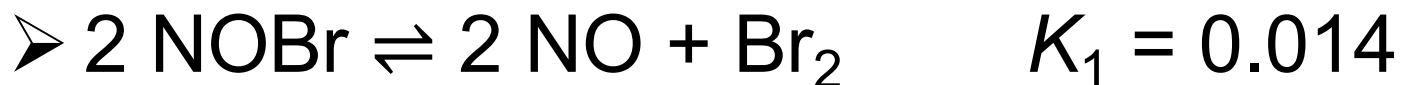
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Equilibria

Consecutive Equilibria

- When two consecutive equilibria occur, the equations can be added to give a single equilibrium.
- The equilibrium constant of the new reaction is the *product* of the two constants:

$$K_3 = K_1 \times K_2$$

- Example



$$K_3 = K_1 \times K_2 = 0.014 \times 7.2 = 0.10$$

Br₂ = 1.0 M conc

$$K_1 = \frac{[\text{NO}]^2 [\cancel{\text{Br}_2}]}{[\text{NOBr}]^2}, \quad K_2 = \frac{[\text{BrCl}]^2}{[\cancel{\text{Br}_2}] [\text{Cl}_2]} \rightarrow K_3 = \frac{[\text{BrCl}]^2 [\text{NO}]^2}{[\text{NOBr}]^2 [\text{Cl}_2]} = K_1 \times K_2$$

Homogeneous vs. Heterogeneous

- **Homogeneous equilibria** occur when all reactants and products are in the same phase.
- **Heterogeneous equilibria** occur when something in the equilibrium is in a different phase.
- The value used for the concentration of a pure substance is always 1.



The Decomposition of CaCO_3 — A Heterogeneous Equilibrium

- The equation for the reaction is

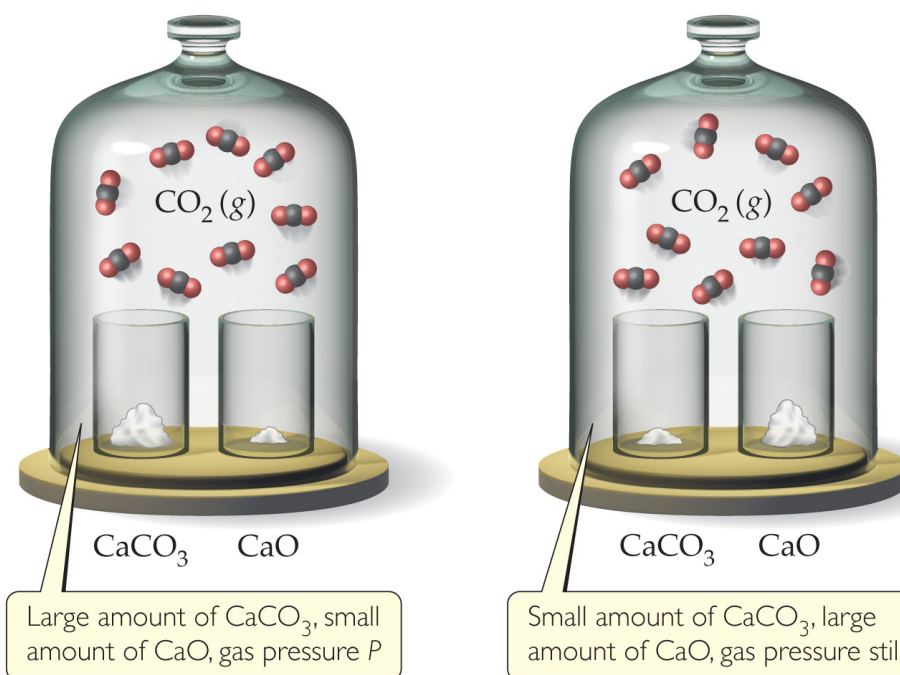
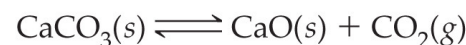


- This results in

$$K_c = [\text{CO}_2]$$

and

$$K_p = P_{\text{CO}_2}$$



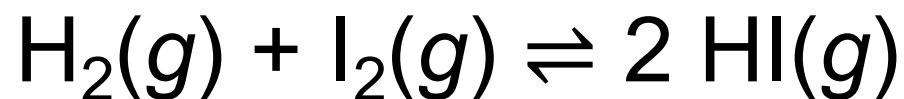
Deducing Equilibrium Concentrations

- 1) Tabulate all known initial and equilibrium concentrations.
- 2) For anything for which initial *and* equilibrium concentrations are known, calculate the change.
- 3) Use the balanced equation to find change for all other reactants and products.
- 4) Use initial concentrations and changes to find equilibrium concentration of all species.
- 5) Calculate the equilibrium constant using the equilibrium concentrations.



An Example

A closed system initially containing $1.000 \times 10^{-3} \text{ M H}_2$ and $2.000 \times 10^{-3} \text{ M I}_2$ at 448° C is allowed to reach equilibrium. Analysis of the equilibrium mixture shows that the concentration of HI is $1.87 \times 10^{-3} \text{ M}$. Calculate K_c at 448° C for the reaction taking place, which is



What Do We Know?

	$[\text{H}_2], M$	$[\text{I}_2], M$	$[\text{HI}], M$
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change	-0.935×10^{-3}	-0.935×10^{-3}	
At equilibrium	6.5×10^{-5}	1.065×10^{-3}	1.87×10^{-3}



[HI] Increases by $1.87 \times 10^{-3} \text{ M}$

	[H ₂], <i>M</i>	[I ₂], <i>M</i>	[HI], <i>M</i>
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change			$+1.87 \times 10^{-3}$
At equilibrium			1.87×10^{-3}



Stoichiometry tells us $[\text{H}_2]$ and $[\text{I}_2]$ decrease by half as much.

	$[\text{H}_2], M$	$[\text{I}_2], M$	$[\text{HI}], M$
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change	-9.35×10^{-4}	-9.35×10^{-4}	$+1.87 \times 10^{-3}$
At equilibrium			1.87×10^{-3}



We can now calculate the equilibrium concentrations of all three compounds.

	$[\text{H}_2], M$	$[\text{I}_2], M$	$[\text{HI}], M$
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change	-9.35×10^{-4}	-9.35×10^{-4}	$+1.87 \times 10^{-3}$
At equilibrium	6.5×10^{-5}	1.065×10^{-3}	1.87×10^{-3}



And, therefore, the equilibrium constant...

$$\begin{aligned}K_c &= \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} \\&= \frac{(1.87 \times 10^{-3})^2}{(6.5 \times 10^{-5})(1.065 \times 10^{-3})} \\&= 51\end{aligned}$$

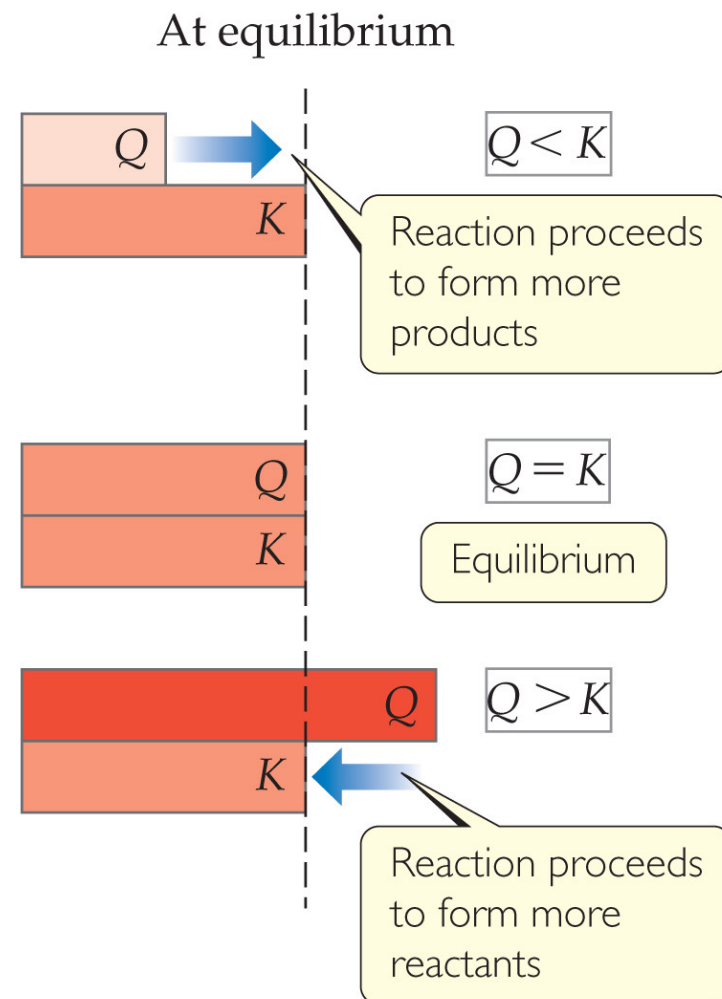


Is a Mixture in Equilibrium? Which Way Does the Reaction Go?

- To answer these questions, we calculate the ^{non-equilibrium} **reaction quotient, Q** .
- Q looks like the equilibrium constant, K , but the values used to calculate it are the **current conditions, not necessarily those for equilibrium.**
- To calculate Q , one substitutes the initial concentrations of reactants and products into the equilibrium expression.

Comparing Q and K

- Nature wants $Q = K$.
- If $Q < K$, nature will make the reaction proceed to products.
- If $Q = K$, the reaction is in equilibrium.
- If $Q > K$, nature will make the reaction proceed to reactants.



Aqueous
Equilibria

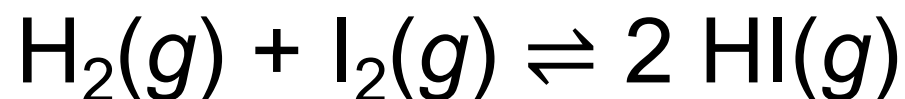
Calculating Equilibrium Concentrations

- If you know the equilibrium constant, you can find equilibrium concentrations from initial concentrations and changes (based on stoichiometry).
- You will set up a table similar to the ones used to find the equilibrium concentration, but the “change in concentration” row will simple be a factor of “x” based on the stoichiometry.



An Example

A 1.000 L flask is filled with 1.000 mol of $\text{H}_2(g)$ and 2.000 mol of $\text{I}_2(g)$ at 448 °C. Given a K_c of 50.5 at 448 °C, what are the equilibrium concentrations of H_2 , I_2 , and HI ?



initial concentration (M)	1.000	2.000	0
change in concentration (M)	$-x$	$-x$	$+2x$
equilibrium concentration (M)	$1.000 - x$	$2.000 - x$	$2x$

Example (continued)

- Set up the equilibrium constant expression, filling in equilibrium concentrations from the table.

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(2x)^2}{(1.000 - x)(2.000 - x)} = 50.5$$

- Solving for x is done using the quadratic formula, resulting in $x = 2.323$ or 0.935 .



Example (completed)

- Since x must be subtracted from 1.000 M , 2.323 makes no physical sense. (It results in a negative concentration!) The value *must* be 0.935 .
- So
- $[\text{H}_2]_{\text{eq}} = 1.000 - 0.935 = 0.065\text{ M}$
- $[\text{I}_2]_{\text{eq}} = 2.000 - 0.935 = 1.065\text{ M}$
- $[\text{HI}]_{\text{eq}} = 2(0.935) = 1.87\text{ M}$



LeChâtelier's Principle

“If a system at equilibrium is disturbed by a change in temperature, pressure, or the concentration of one of the components, the system will shift its equilibrium position so as to counteract the effect of the disturbance.”



How Conditions Change Equilibrium

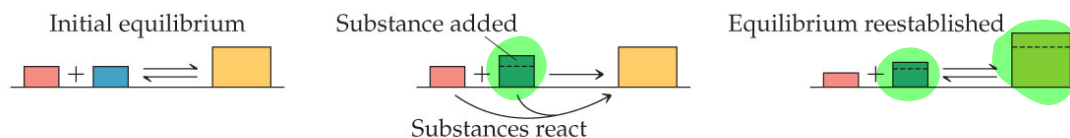
We will use LeChâtelier's Principle qualitatively to predict shifts in equilibrium based on changes in conditions.

Le Châtelier's Principle

If a system at equilibrium is disturbed by a change in **concentration**, **pressure**, or **temperature**, the system will shift its equilibrium position so as to counter the effect of the disturbance.

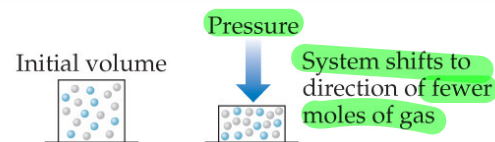
Concentration: adding or removing a reactant or product

If a substance is added to a system at equilibrium, the system reacts to consume some of the substance. If a substance is removed from a system, the system reacts to produce more of substance.



Pressure: changing the pressure by changing the volume

At constant temperature, **reducing the volume of a gaseous equilibrium mixture causes the system to shift in the direction that reduces the number of moles of gas.**



Temperature:

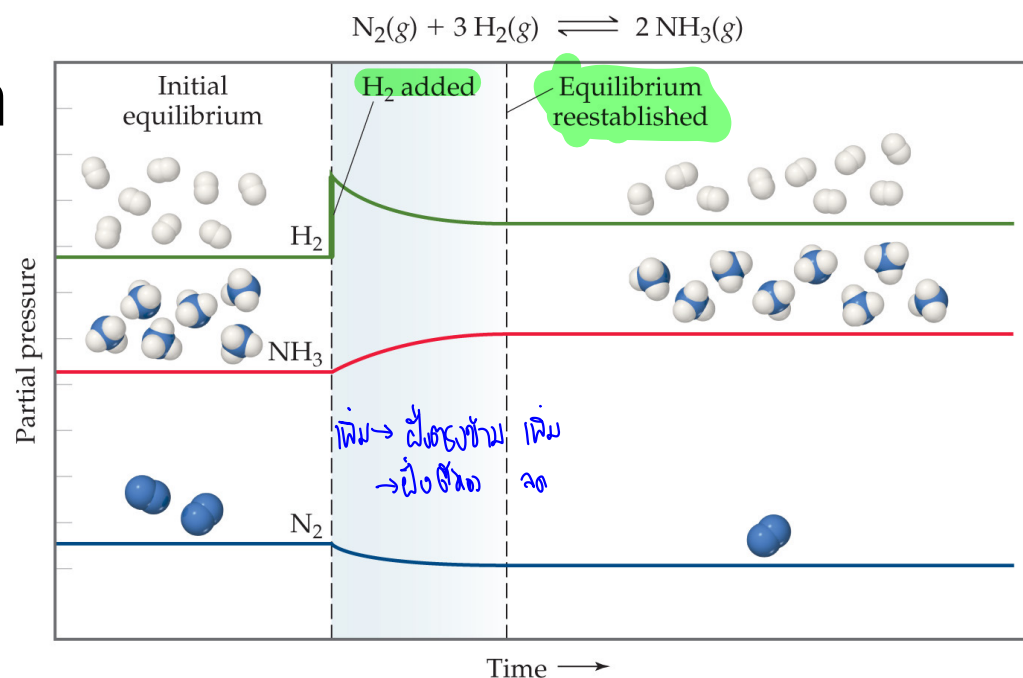
If the **temperature of a system at equilibrium is increased**, the system reacts as if we added a **reactant to an endothermic reaction** or a **product to an exothermic reaction**. The equilibrium shifts in the **direction that consumes the "excess reactant,"** namely heat.



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Change in Reactant or Product Concentration

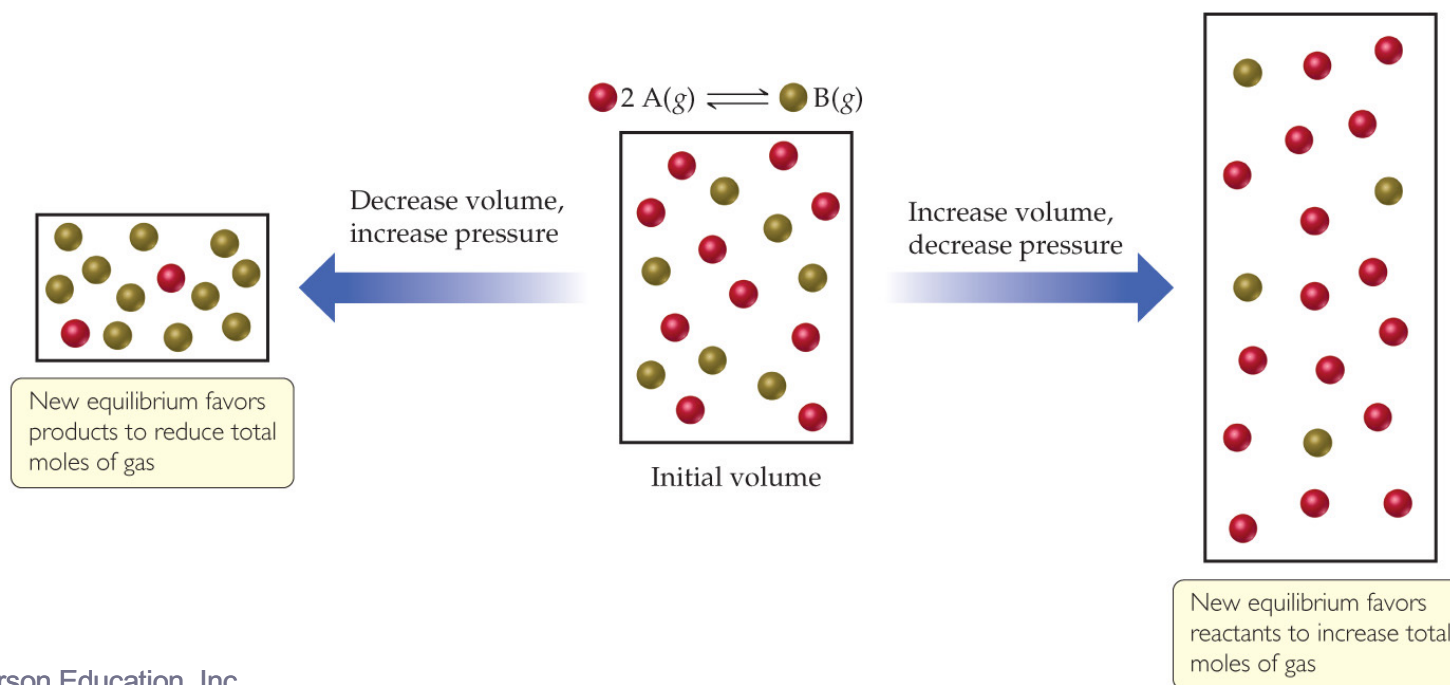
- If the system is in equilibrium
 - adding a reaction component will result in some of it being used up.
 - removing a reaction component will result in some of it being produced.



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Change in Volume or Pressure

- When gases are involved in an equilibrium, a change in pressure or volume will affect equilibrium:
 - Higher volume or lower pressure favors the side of the equation with more moles (and

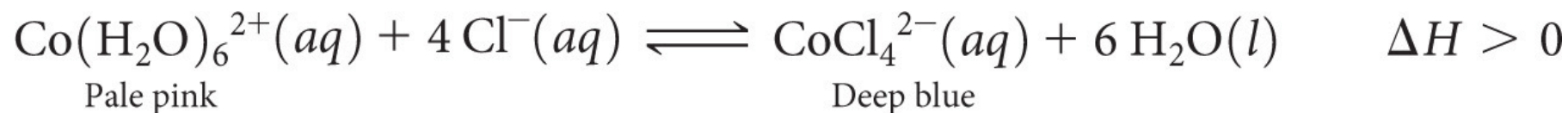


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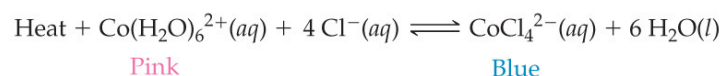
Change in Temperature

- Is the reaction endothermic or exothermic as written? That matters!
- Endothermic: Heat acts *like* a reactant; adding heat drives a reaction toward products.
- Exothermic: Heat acts *like* a product; adding heat drives a reaction toward reactants.

An Endothermic Equilibrium



$\Delta H > 0$, endothermic reaction



The diagram illustrates the endothermic equilibrium between two cobalt complexes in aqueous solution. It features three beakers connected by arrows indicating temperature changes:

- Left Beaker (Cooling):** A blue arrow labeled "Cool" points from the middle beaker to this one. The solution is pink. A circular inset above shows a high concentration of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions and a low concentration of blue CoCl_4^{2-} ions.
- Middle Beaker (Equilibrium):** The solution is violet. A circular inset above shows a mixture of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and blue CoCl_4^{2-} ions.
- Right Beaker (Heating):** A red arrow labeled "Heat" points from the middle beaker to this one. The solution is blue. A circular inset above shows a high concentration of blue CoCl_4^{2-} ions and a low concentration of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions.

Explanatory text boxes for each beaker:

- Left Beaker:** Solution appears pink because lowering the temperature shifts the equilibrium to favor formation of the pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ion.
- Middle Beaker:** Solution appears violet because appreciable amounts of both pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and blue CoCl_4^{2-} are present.
- Right Beaker:** Solution appears blue because raising the temperature shifts the equilibrium to favor formation of the blue CoCl_4^{2-} ion.

Aqueous
Equilibria

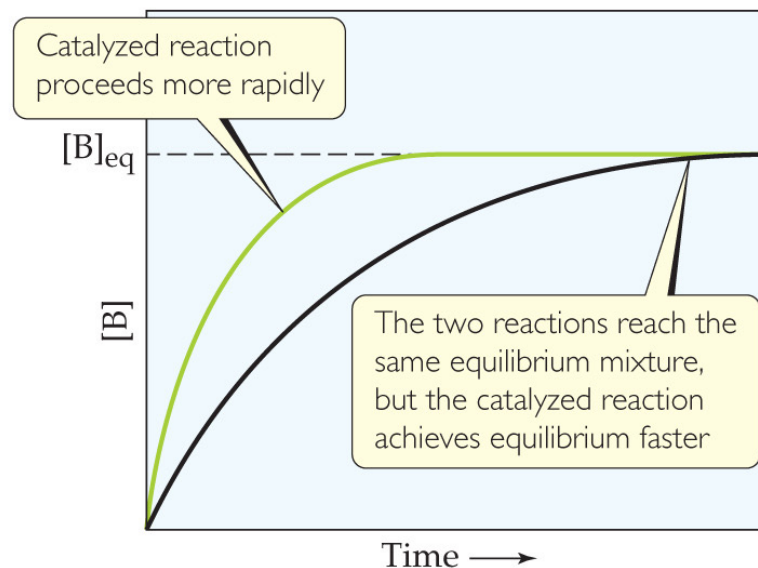
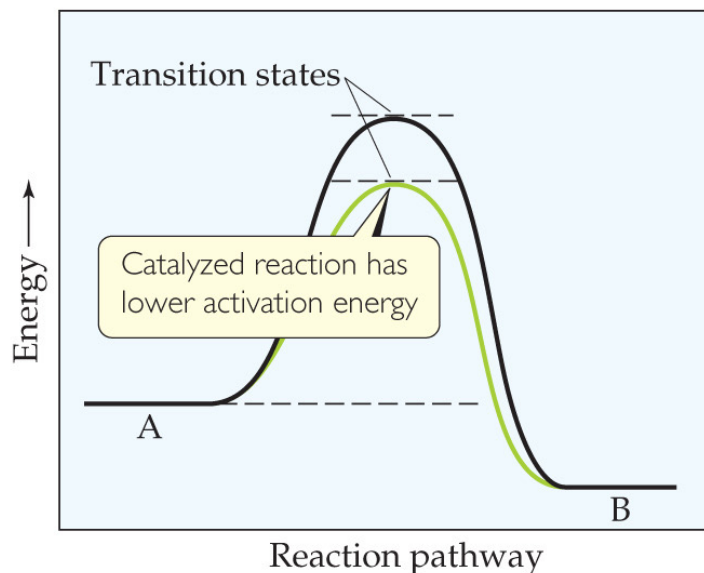
An Exothermic Equilibrium

- The Haber Process for producing ammonia from the elements is exothermic.
- One would think that cooling down the reactants would result in more product.
- However, the activation energy for this reaction is high!
- This is the *one* instance where a system in equilibrium can be affected by a catalyst!



Catalysts

- **Catalysts** increase the rate of both the forward *and* reverse reactions.
- Equilibrium is achieved faster, but the equilibrium composition remains unaltered.
- Activation energy is lowered, allowing equilibrium to be established at lower temperatures.



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